# VALENTYNA KUNDAS

Microbiologist

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## EDUCATION

### Bachelor's degree in Biotechnology

Taras Shevchenko National University of Kyiv, Ukraine
Final grade: 4.65/5 (3.7 GPA) *09/2018 – 06/2022*
*Courses:* Basics of Biotechnology, Biochemistry, Microbial synthesis, Plant Biotechnology, Molecular biology
*Extra:* Antimicrobial resistance, Python programming

### Master's degree in Molecular Bioengineering

Technische Universität Dresden, Germany
Final grade: 3.8 GPA *10/2022 – 09/2024*
*Courses:* Biostatistics, Biophysics, Genomes and Evolution, Proteomics, Bioimage analysis, Biomedicine
*Extra:* Algae Biotechnology

### PhD in Environmental Science

*11/2024 – present*
University of Toronto Scarborough, Canada
Current grade: 3.95 GPA
Dr. Terrence Bell's Microbiome Manipulation Lab since September 2025
**Research focus:** plant-bacteria interactions, environmental stress, and sustainable agriculture.
*Courses:* Microorganisms and the Environment, Applied Biogeochemistry and Geomicrobiology

## RESEARCH EXPERIENCE

### PhD Research – Alfalfa Rhizosphere Microbiome

*(09/2025 – present)*
Department of Physical and Environmental Sciences, University of Toronto Scarborough, Scarborough, Canada

- My research investigates how combinations of environmental conditions influence soil microbiome composition and function. I study plant-bacteria interactions in soil using Alfalfa (*Medicago sativa*) as a model plant. The work focuses on how plant stress and environmental factors such as light and water availability shape microbiome in soil.
- Developed a four-generation iterative microbiome-selection framework under repeated drought and am currently evaluating microbiome and plant responses using 16S rRNA gene sequencing and phenotyping.
- Designed and am conducting home-home/home-away swap experiments to test whether drought-selected microbiome responses persist when water-stress conditions change.
- Integrating microbiome and plant-trait data to identify bacterial patterns associated with biomass, root characteristics, and leaf greenness.

### Alfalfa-Microbiome Simulation Model

*(05/2026 – present)*
Department of Physical and Environmental Sciences, University of Toronto Scarborough, Scarborough, Canada

- Developed an interactive online tool to explore alfalfa responses under varying environmental and plant-microbiome conditions. The tool is being refined using experimental data from my PhD research.
  *Project: Phytobiome* https://av-bio.github.io/phytobiome/

### Algae Biotechnology Master's thesis

*(03/2024 – 08/2024)*
GEPEA lab & AlgoSolis platform & Nantes University, Saint-Nazaire, France

- The project aimed to scale up and adjust conditions for the creation of a calcite-based protective coating on the surface of concrete using microalgae to protect wind turbines in a marine environment. Multiple conditions were tested to enhance biofilm formation and biomineralization on the surface of concrete.
- Explored enhanced calcite deposition using microalgae compared to abiotic spontaneous precipitation.

### Algae Biorefinery project

*(09/2023 – 03/2024)*
Institute of Natural Materials Technology, Dresden University of Technology, Dresden, Germany

- Discovered the most favorable conditions for pigment production, including astaxanthin, under green, white, blue, and red-light spectra. Techniques: continuous microalgae cultivation using CellDeg photobioreactor, lipid extraction, pigment extraction, dry biomass, spectrophotometry.
- Project: "Analysis of the macromolecular composition of microalgae as a function of spectral light condition."

### Plant genetic internship

*(04/2022 – 09/2022)*
Chair of Plant Cell and Molecular Biology, Dresden University of Technology, Dresden, Germany

- Plant genomics and cytogenetics, focusing on repetitive DNA. Techniques: genomic DNA extraction and purification, PCR, gel electrophoresis, linear DNA digestion, and rolling circle amplification.

**Plant biotechnologist** *(02/2021 – 02/2022)*
Institute of Cell Biology and Genetic Engineering, National Academy of Science of Ukraine
- Implemented plant reproduction *in vitro,* optimized methodology of *Lemna spp* sterilization, cultivation, *Agrobacterium*-mediated genetic transformation.
- Worked in biosafety cabinets with a vertical and a horizontal laminar flow hood, prepared solid media.
- Thesis: "The use of *Lemnaceae* representatives in biotechnology and environmental protection."

**Microbial biochemistry internship** *(10/2019 – 02/2022)*
Zabolotny Institute of Microbiology and Virology, The National Academy of Science of Ukraine
- Experienced in β-glucans and lipopolysaccharide (LPS) extraction, identifying carbohydrates, proteins, nucleic acids. Term paper "*Azospirillum brasilense* β-glucans role in the diazotroph-plant associative system."

**Molecular biology internship** *(07/2019 – 03/2020)*
Diagen Genetic Laboratory, Kyiv, Ukraine
- Experienced in DNA isolation, gel electrophoresis, PCR, statistical and data analysis for the project "Substance abuse genetics: in search for shared and substance-specific variants."

## AWARDS

- Yorkshire Valley Farms Canadian Eco-Scholar Award Recipient – a scholarship recognizing the project working on organic regenerative agriculture *(06/2025)*
- Deutschlandstipendium holder – scholarship for talented and high-achieving students *(10/2022-09/2023)*
- Ukrainian State Scholarship students awarded to 40% of the specialty *(09/2018 – 06/2022)*
- I finished middle and high school with honors.

## PEER-REVIEW

- Co-reviewer, *New Phytologist*, 2026

## RESEARCH PRESENTATIONS

**Kundas, V.**, & Bell, T. H. *Environmental and Host-Mediated Selection of Alfalfa Rhizosphere Microbiomes Across Successive Microbial Generations Under Drought.* International Phytobiomes Conference 2026, Niagara-on-the-Lake, Canada. Poster.

**Kundas, V.**, & Bell, T. H. *Environmental Stress Shapes Plant-Associated Soil Microbiomes.* Graduate Cohort Research Presentation, University of Toronto Scarborough, Toronto, Canada, 2026. Oral presentation.

**Kundas, V.**, Renaudie, M., & Hilloulin, B. *Creation of a Protective Coating for Structures in a Marine Environment: Precipitation of Calcite Through Microalgae on the Surface of Cementitious Materials.*

- Workshop on Biominerals at University of Tübingen, Germany, 2025. Poster.
- University of Toronto Scarborough, Toronto, Canada, 2025. Oral presentation.
- GEPEA, Nantes, France, 2024. Oral presentation.
- Technische Universität Dresden, Dresden, Germany, 2024. Oral presentation.

**Kundas, V.**, Krujatz, F., & Bleisch, R. *Analysis of the Macromolecular Composition of Microalgae as a Function of Spectral Light Condition.* Departmental Research Presentation, Technische Universität Dresden, Dresden, Germany, 2024. Oral presentation.

## PROFESSIONAL DEVELOPMENT AND WORKSHOPS

- **Workshop on Biominerals** at University of Tübingen, Germany: synthesis of biogenic and abiotic minerals; Mössbauer spectroscopy, XRD, SEM, quantification of Fe species, magnetic susceptibility, surface charge and particle size analysis, thin section microscopy, microscopy of bacteria with mineral inclusions *(07/2025)*
- **Summer school "Biomarkers for Lakes Ecological Monitoring"** at the UMR CARRTEL (INRAE-USMB), Thonon-les-Bains, France: diagnosis of ecological status of the lakes and rivers, specific diversity (eDNA, zooscan), ecosystem functioning and biotic interactions (the use of stable isotopes as biomarkers) *(06/2025)*

## TEACHING AND MENTORSHIP EXPERIENCE

**Teaching assistant:**

2025: BIOB32H3 Animal Physiology Laboratory
EES1123H3 Environmental Regulations
CHMB62H3 Introduction to Biochemistry
EESA11H3 Environmental Pollution
Summer School: Air Quality Science
CHMA10H3 Introductory Chemistry I: Structure and Bonding

2026: EES1123H Environmental Regulations (returning)
BIOC17H3 Microbiology
EESA11H3 Environmental Pollution (returning)
BIOC17H3 Microbiology (returning)
Summer School: Solar panels
Summer School: Chemistry
CHMA10H3 Introductory Chemistry I: Structure and Bonding (returning)

Most responsibilities include leading a lab for 24 students, grading assignments, conducting tutorials for up to 40 students, giving feedback, and communicating with students.

**Mentorship:**

- Supervised an undergraduate researcher developing an *in vitro* sterilization protocol for *Medicago sativa* seedlings; guided experimental design, laboratory techniques, troubleshooting, data analysis, and interpretation *(07/2026–09/2026)*
- Mentored an independent undergraduate research project on iterative microbiome passaging in liquid culture, providing guidance on experimental design, technical training, troubleshooting, and interpretation of results *(05/2026–present)*

I have received numerous emails from both students and professors recognizing my contributions and performance as a teaching assistant.

## ACADEMIC LEADERSHIP AND SERVICE

- **Organized and facilitated an online conference** for 40 graduate students enrolled in the Environmental Pollution course *(04/2026)*
- **Student body involvement:** Scarborough Campus director at University of Toronto Graduate Student Union *(05/2025-10/2026)*; Physical and Environmental Sciences Representative at the Graduate Student Association at Scarborough *(09/2025-08/2026);* Secretary of the Institute Student Parliament, an active member of the University and Institute Student Parliament, Class leader *(10/2018 – 06/2022)*
- **Volunteering:** member of “Bionity” (promotion and popularization of science), participant of "Biology - the science of the future" (disseminating in-depth knowledge of biology among school students), part of "Iron Volunteers" team (organizations of sports activities for charity) *(05/2017 – 06/2022)*
- **Acknowledged with the Rector honors**, the top 10 most dedicated participants from the Institute for their exceptional contributions as entrants' consultants, assisting the Institute of Biology and Medicine Admissions Committee *(06/2019-08/2021)*

## SKILLS

**Biological skills**: light microscopy, SEM, X-Ray diffraction, dry matter, microalgae cultivation, spectrophotometry, pigment extraction, work in biosafety cabinets, reproduction of plants *in vitro*, plant transformation, seed bacterial cell culture, preparation of growth media, DNA isolation, PCR, gel electrophoresis, histology sample preparation, thermogravimetric analysis, plant growth experiments.
**Computer skills**: Python, Microsoft Office pack, R software, WinRHIZO, Resonon.
**Soft skills:** adaptability, teamwork, leadership, multitasking.

## LANGUAGES

**English** – advanced **Ukrainian** – native **German** – beginner